Ericsson PE Node (BAS-PE-10) Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) Resolution SOP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_4358

Description

This Standard Operating Procedure (SOP) addresses the failure domain of hardware/environmental faults on the Ericsson PE Node (BAS-PE-10) at the Basavanagudi POP, specifically focusing on ASIC memory parity errors, line card crashes, and power supply redundancy loss. The PE Node, responsible for providing a high-capacity and reliable transport network, plays a critical role in ensuring the integrity of the telecommunication network. However, hardware-related issues can significantly impact the network's performance and availability. ASIC memory parity errors occur when the memory management unit detects an error in the data stored in the Application-Specific Integrated Circuit (ASIC), while line card crashes are caused by a malfunctioning line card that handles the transmission and reception of data. Power supply redundancy loss, on the other hand, is triggered when the redundant power supply fails to take over in case of a primary power supply failure. These issues are typically detected by the alarm code ALARM_4358, which has a severity level of Major. To address these issues, it is essential to identify the root cause and take corrective actions to prevent further damage to the network infrastructure.

Pre-requisite Checks

- Ensure that the Basavanagudi POP is accessible and that remote hands are dispatched to the site.

- Verify the status of the redundant link to ensure that it is operational and can take over in case of a primary link failure.

- Confirm the power supply redundancy status to ensure that both power supplies are functioning correctly.

- Perform a thorough visual inspection of the PE Node to identify any signs of physical damage or environmental issues.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to faulty memory	High
Line card crash due to hardware malfunction	Medium
Power supply redundancy loss due to faulty power supply	Low

Diagnostic Commands & Expected Outputs

```
Command: show node info, Expected Output: Node ID: BAS-PE-10, Node Type: PE, Software Version: 10.3, Description: This command displays the node information, including the node ID, type, and software version. This helps to verify that the node is functioning correctly and that the software is up-to-date.

Command: show alarm history, Expected Output: ALARM_4358: Major, Time: 2023-02-20 14:30:00, Description: ASIC memory parity error, Description: This command displays the alarm history, including the alarm code, severity, time, and description. This helps to identify the root cause of the issue and determine the correct course of action.

Command: show line card status, Expected Output: Line Card 1: Operational, Line Card 2: Not Operational, Description: This command displays the status of the line cards, including their operational status. This helps to identify if any line cards are malfunctioning and need to be replaced.

Command: show power supply status, Expected Output: Power Supply 1: Operational, Power Supply 2: Operational, Description: This command displays the status of the power supplies, including their operational status. This helps to verify that both power supplies are functioning correctly and can take over in case of a primary power supply failure.
```

Resolution Steps

Step 1: Step: Replace faulty memory or line card, Command: replace node component memory, Description: This step involves replacing the faulty memory or line card with a new one. This can be done using the 'replace node component memory' command, which allows for the replacement of individual components within the node.

Step 2: Step: Replace faulty power supply, Command: replace node power supply, Description: This step involves replacing the faulty power supply with a new one. This can be done using the 'replace node power supply' command, which allows for the replacement of the power supply within the node.

Step 3: Step: Perform a software upgrade, Command: software upgrade, Description: This step involves performing a software upgrade on the node to ensure that it is running the latest software version. This can be done using the 'software upgrade' command, which allows for the upgrade of the software within the node.

Post-Check Verification

```
Command: show node info, Expected Output: Node ID: BAS-PE-10, Node Type: PE, Software Version: 10.3, Description: This command is used to verify that the node is functioning correctly and that the software is up-to-date.

Command: show alarm history, Expected Output: No alarms, Description: This command is used to verify that the alarm has been cleared and that there are no more issues with the node.

Command: show line card status, Expected Output: Line Card 1: Operational, Line Card 2: Operational, Description: This command is used to verify that all line cards are operational and functioning correctly.

Command: show power supply status, Expected Output: Power Supply 1: Operational, Power Supply 2: Operational, Description: This command is used to verify that both power supplies are functioning correctly and can take over in case of a primary power supply failure.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: config restore, Description: This step involves restoring the original configuration of the node to its previous state. This can be done using the 'config restore' command, which allows for the restoration of the configuration within the node.

- Step: Revert the software upgrade, Command: software downgrade, Description: This step involves reverting the software upgrade to its previous version. This can be done using the 'software downgrade' command, which allows for the downgrade of the software within the node.

- Step: Replace the faulty component, Command: replace node component memory, Description: This step involves replacing the faulty component with a new one. This can be done using the 'replace node component memory' command, which allows for the replacement of individual components within the node.